

Sai Manikanta Karnati

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of software engineering building secure, scalable software solutions, modern web applications, and Flutter/Dart mobile applications across healthcare-grade and medical device systems. Skilled in TypeScript, JavaScript, Python, React, CI/CD, and DevOps with strong product management, risk management, and quality assurance instincts. Delivered HIPAA-conscious backend data services that cut issue resolution time 40% and boosted performance 35%.

SKILLS

React - TypeScript - JavaScript - Python - Flutter - Dart - Devops - CI/CD - Node.js - Quality Assurance - Risk Management - Product Management - Hipaa - Medical Device - Software Engineering - Microservices - Spring Boot - Kubernetes - Docker - AWS - GraphQL - REST - SQL - Apache Kafka

EXPERIENCE

Software Engineer - Vanguard

2025 - present

- Architected secure, scalable backend data services in Python and Spring Boot microservices, enabling independent deployments across complex healthcare integrations.
- Engineered high-throughput, event-driven systems with DynamoDB Streams, Python/Node.js Lambda, and SNS/SQS, achieving 99.99% reliability for medical device-grade data processing.
- Built cross-platform Flutter and Dart mobile applications with reusable components, improving customer-facing clinical workflow delivery speed ~30%.
- Developed responsive modern web applications and React/TypeScript components, accelerating frontend delivery for clinical data and Nidra Sync-style synchronization workflows.
- Engineered RESTful and GraphQL APIs in TypeScript and JavaScript, ensuring secure frontend-backend integration across healthcare systems.
- Optimized SQL queries, indexing, and schema design across relational and NoSQL databases, improving reporting and transaction performance 35%.
- Implemented CI/CD and DevOps observability with Splunk, applying quality assurance practices that cut production issue resolution time 40%.
- Led technical design reviews and risk management discussions with engineers, strengthening communication, leadership, product management, and HIPAA-conscious security design.

Associate Software Engineer - Xebia

2022 - 2023

- Engineered backend data services in Python, Spring Boot, and Node.js, delivering scalable RESTful APIs and secure software solutions for enterprise customers.
- Built customer-facing web applications and Flutter/Dart mobile applications using React and TypeScript for complex UI and data synchronization workflows.
- Optimized SQL queries, joins, views, and stored procedures, improving API response times 35% for high-volume data processing.
- Created and documented secure RESTful and GraphQL APIs with Swagger/OpenAPI, enforcing versioning and backward compatibility across integrations.
- Implemented event-driven architecture with Apache Kafka, reducing inter-service dependency failures 35% across system integrations.
- Containerized applications with Docker and Kubernetes and CI/CD pipelines, applying DevOps practices that improved deployment efficiency ~30%.
- Integrated OAuth2, JWT, and Spring Security, hardening secure access for HIPAA-aligned data processing and risk management.

Engineer Trainee - Xebia

2021 - 2021

- Developed enterprise web applications with React/TypeScript frontends and Python/Java backend services, applying solid software engineering practices.

- Optimized SQL queries, stored procedures, and relational schemas for transactional systems, improving query execution time 20%.
- Engineered RESTful APIs and documented service contracts with Swagger/OpenAPI for technical clarity across engineers.
- Implemented API security with OAuth2 and Spring Security plus rate limiting and gateway protections for secure backend services.
- Increased unit test coverage to 95% with JUnit and Python test automation, embedding quality assurance into software development.

EDUCATION

Master of Science in Computer Science - University Of North Carolina At Charlotte

CERTIFICATIONS

- AWS Certified Developer Associate
- MongoDB Certified Developer
- Infosys Certified Software Programmer
- NPTEL Certified Java Developer
- AWS Academy Graduate - Data Engineering